Clear Chain-of-Thought (ClearCoT)

Philosophical Clarity as an Optimizer Tool

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Abstract

While Chain-of-Thought (CoT) prompting has dramatically improved the reasoning capabilities of Large Language Models (LLMs), it introduces severe computational inefficiencies through excessive token generation. Two dominant failure patterns are detected: "Token Wall" failures (endless generative loops of self-doubt and conversational filler) and overly linear step-by-step or dotted-checklist reasoning before reaching a conclusion. To address this, Clear Chain-of-Thought (ClearCoT) is introduced, a methodology based on the philosophical principle of "clear and precise thinking" that enforces a rigorous structural connection between the Question, Reasoning process, and Answer (Q–R–A).

A novel perspective tested and executed entirely on consumer-grade local hardware (24 GB VRAM). First, a 27B-parameter Teacher model synthesizes philosophical reasoning pairs, which are used to train a 4B-parameter MasterMind Model. This MasterMind subsequently acts as a "logical optimizer" tool, restructuring publicly available multi-domain datasets into highly logically optimized examples used to fine-tune a 2B-parameter test model (Occam-2B).

Evaluated against the base model, Occam-2B demonstrates a statistically significant 6.71 % accuracy increase on MMLU-Pro ($P < 0.0001$) while drastically reducing 95th-percentile token bloat by approximately 69 %. On complex logic tasks, the ClearCoT methodology eliminates up to 89.4 % of reasoning tokens and almost entirely resolves catastrophic "Token Wall" failure loops. Ultimately, this paper demonstrates that prioritizing logical structure over data volume enables small models to achieve reasoning precision, establishing the MasterMind Model as a highly effective, reusable tool for optimizing any available dataset featuring a `<think>` process. The resulting Occam-2B model is publicly available at https://huggingface.co/collections/Sagicc/occam-clearcot.

1 Introduction

In nature, a tree shows its beautiful, expansive, and complex canopy, but in the context of AI reasoning this beautiful complexity generates problems. Standard Chain-of-Thought (CoT) reasoning models mimic human reasoning processes by generating extensive conversational filler that resembles a dense canopy. When faced with complex problems, these models output long internal dialogues, debating alternatives that are mostly nonsense. While this "canopy-like" reasoning appears natural (which it often is not), it demands massive computational resources, extends GPU runtime, and significantly increases inference costs [1, 5].

1.1 Philosophy in ClearCoT

Analysis of regular Chain-of-Thought processes generated by contemporary LLMs uncovers two dominant reasoning patterns. The first is expansive and tree-like, characterized by wide branching loops ("Wait, the user asks…", self-doubt, and conversational filler). The second is compressed into short checklists or linear step-by-step sequences. Both patterns rest on statistical likelihood rather than structural clarity.

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Clear Chain-of-Thought (ClearCoT) adopts an entirely different strategy: it analyzes the complete path from **Question** through **Reasoning** to **Answer** and enforces their meaningful connections. Rather than engineering prompts or compressing post-hoc, this methodology starts by distilling philosophical clarity directly into the model.

1.2 The MasterMind Model (MMM)

The basic task was to create a specialized "MasterMind Model" (MMM), a **logical optimizer**, with the assignment to "reconstruct" the reasoning procedure. This model serves as a tool for logical optimization of available datasets to be used for further fine-tuning.

In this paper, it is demonstrated that prioritizing logical and meaningful structure over data volume enables even a small model to fundamentally rewire its parametric behaviour and achieve reasoning precision. To test this method of ClearCoT, the small language model Qwen3.5-2B[1] is selected, SFT fine-tuned, and named Occam-2B. Quantitative evaluations show that Occam-2B not only eliminates the Token Wall failure state but also delivers higher accuracy on challenging benchmarks such as MMLU-Pro (6.71 %), maintains PhD-level reasoning on GPQA Diamond, and strips away roughly 64 % of wasted reasoning tokens.

Appendix contains datasets, chat, and reasoning examples.

2 Methodology

The methodology developed in this experiment follows the classical philosophical steps of organizing research:

detect and define a problem → analyse it → develop possible solution

Regular Chain-of-Thought (CoT) can lead to a failure state named the "Token Wall" death-loop: a phenomenon where a model becomes trapped in endless self-debates, generating conversational filler and self-doubt until it exhausts its maximum context window and completely fails to reach a final answer. Another detected feature is linear step-by-step and dotted-list reasoning that only reaches an answer at the very end. To address this, Clear Chain-of-Thought (ClearCoT) is proposed as a paradigm shift from mechanical prompt engineering to intrinsic philosophical alignment. Based on the philosophical principle of **"clear and precise thinking"** that establishes a clear relation between Question, Reasoning, and Answer, ClearCoT trains models to begin reasoning with immediate logical structure and to conclude with a well-formatted final answer. While recent literature emphasizes large-scale synthetic data generation [7, 4] or cascading Teacher-Student distillation pipelines [2], ClearCoT offers an alternative, independently developed strategy that connects all Q-R-A components. with a focus on the Thinking mechanism (Reasoning), which requires logical optimization and **Awareness**[2]. For this purpose, a new MasterMind Model was developed that systematically strips away conversational canopy, extracts the meaningful path to the answer, and rewrites datasets into clear and precise examples.

2.1 Phase I: Foundational Logic Optimization (The 27B Teacher)

To establish the baseline for the Clear Chain-of-Thought (ClearCoT) methodology, the first requirement was the creation of a dataset strictly based on philosophy and logic. The Stanford Encyclopedia of Philosophy dataset (AiresPucrs/stanford-encyclopedia-philosophy)[3] was an appropriate starting point. However, it was necessary to clean and merge articles and chunk them into coherent meaningful units. For each unit, two parallel `<think>` trajectories were generated: a verbose (large) version of typical LLM reasoning and a clear, shorter version. New metadata "principles applied" was appended to explicitly declare the underlying

[1] https://huggingface.co/Qwen/Qwen3.5-2B

[2] To be aware of both Q and A while performing R

[3] https://huggingface.co/datasets/AiresPucrs/stanford-encyclopedia-philosophy

philosophical problems (e.g., Essential Sameness, Difference in Property, Ontological Necessity). View the full example in Appendix A.1.

The Qwen3.5-27B model was selected as the primary Teacher. Since hardware capacity limits the use of the full model, a GGUF version from Ollama[4] was used. This 27B model was tasked to process philosophical units sourced from the prepared dataset.

A system prompt for Student:

"You are a logical optimizer. Analyse the reasoning and output a rigorous, shorter <think> block."

```
prompt = f"""You are an expert data generation assistant. Your task is to analyze academic text and output STRICTLY valid JSON.

Read the following large philosophical/scientific text chunk:
---
{chunk_or_problem}
---

{domain_prompt_prefix}This text is dense. Identify the **single most prominent logical argument or principle** within it, and base your training example strictly around that focus.

Instructions for your JSON output:
1. "user_question": A complex, realistic question where this specific principle applies. MUST explicitly include any names or specific entities that will be referenced in the reasoning.
2. "verbose_reasoning": A flawed, rambling, overly long internal monologue that is mostly coherent but contains EXACTLY ONE clear minor flaw (e.g., a circular reference or an unsupported leap from premise to conclusion). Keep it grounded in the chunk. Do NOT use XML or think tags in this string.
3. "optimized_reasoning": A highly compressed, rigorous, flaw-free internal monologue using the exact logical framework from the text. Do NOT use XML or think tags in this string.
4. "final_answer": The concise conclusion.
5. "principles_applied": Optional list of 1-4 short strings naming the exact principles used in the optimized reasoning.

OUTPUT STRICTLY AS A VALID JSON OBJECT matching this schema. Do not output conversational filler or markdown outside the JSON.
Schema for reference:
{json.dumps(schema, indent=2)}
"""
```

The Teacher extracted the underlying principles from chunk units and optimized reasoning trajectories. This phase generated a foundational dataset of approximately 6,000 highly optimized philosophical rows.

For each unit, two parallel <think> trajectories were generated: a verbose version mirroring typical LLM reasoning and a clear, shorter version. New metadata "principles applied" was appended to explicitly declare the underlying philosophical notions (e.g., Essential Sameness, Difference in Property, Ontological Necessity).

The resulting MasterMind Model internalizes this philosophical methodology of "clear and precise thinking" and becomes the reusable optimizer tool for further adaptation of general reasoning datasets (see Sec-

[4]https://ollama.com/library/qwen3.5:27b

tion 2.3). Occam-2B[5] serves merely as the proof of concept; the MasterMind Model is the main innovation.

2.2 Phase II: MasterMind Model (4B)

To integrate a clear philosophical methodology into general reasoning, a tool serving as a "logical optimizer" is required. Supervised Fine-Tuning (SFT) was applied to the selected Qwen3.5-4B[6] model using the ~6,000-row Stanford dataset generated in Phase I, formatted in ChatML. This specific architecture was selected based on three primary criteria: low computational resource demands, an optimal balance between parameter count (4B) and reasoning effectiveness, and an open-source license.

Example: Stanford Dataset MasterMind Adaptation – structure

System: You are a logical optimizer. Analyze the reasoning and output a rigorous, shorter `<think>` block.
User: Problem: According to Abelard's theory of identity in Theologia christiana, how can the Father, Son, and Holy Spirit be three distinct Persons while remaining one God, specifically regarding their essential sameness versus their difference in property?
Verbose Reasoning:`</think>`Okay, so we need to figure out how Abelard solves the Trinity problem using his specific modes of identity. First, he rejects the idea that there is just a plain meaning, [...]. But wait, ...`</think>`
Assistant: `<think>` Abelard resolves the Trinity by distinguishing modes of identity: (1) Essential Sameness: The Father, Son, and Holy Spirit are numerically the same concrete thing (God), making them essentially identical. (2) Difference in Definition: Their specific roles (Fatherhood, Sonship) differ in definition. (3) Difference in Property: They possess unique, 'unmixed' properties ... `</think>`
Principles applied: Essential Sameness, Difference in Property, Unmixed vs. Mixed Properties
Answer: The Persons are essentially the same as one another because they are the numerically identical concrete thing (God), yet they differ in property through unique, unmixed characteristics (e.g., unbegotten, begotten) that distinguish them without creating numerical difference from God.

View full example in Appendix A.1.

This resulting 4B model, named as the MasterMind Model (MMM), successfully internalized the **ClearCoT** and adopted philosophical methodology of "clear and precise thinking". By transferring the 27B's deductive capabilities into a 4B architecture, a highly efficient, specialized "logical optimizer" was created, capable of applying structured philosophical principles to novel, non-philosophical domains.

2.3 Phase III: Dataset Adaptation for General Reasoning

In the third phase, the MasterMind Model (4B) was deployed to perform Dataset Optimization on general and mathematical examples on **nvidia/Nemotron-Cascade-SFT-Stage-1**[7] (2000 rows from "general" column) and **nohurry/Opus-4.6-Reasoning-3000x-filtered**[8] (1000 rows). The MasterMind Model analyzed these generalized tasks and restructured them to match the ClearCoT methodology. It followed the Q–R–A path, analyzed the original `<think>` and generated a new, optimized, `<think>` block with **awareness of Q and A**. Here are the structure examples from both datasets, while complete examples are presented in Appendix.

[5] Model named in regards to William of Occam and his principle that "Must not multiply entities beyond necessity".
[6] https://huggingface.co/Qwen/Qwen3.5-4B
[7] https://huggingface.co/datasets/nvidia/Nemotron-Cascade-SFT-Stage-1
[8] https://huggingface.co/datasets/nohurry/Opus-4.6-Reasoning-3000x-filtered

Example: Opus-4.6-Reasoning - structure

Source: Opus-4.6-Reasoning **principles applied:** Sentiment Analysis via Affirmative Descriptors, Contextual Interpretation of Popularity Metrics **original reasoning:** <think>Analysis of the Cold Stone Creamery Review Step 1: Identify key phrases and sentiment[...]</think>
System: You are a helpful, logical assistant.
User: Cold Stone! Fun and delicious.... It's a great place to bring your kids to let them pick out their sugar introduced comma. This location is always packed because of where it's at but I've never had bad service from here or any of their other locations. Is this review positive or negative? Options: (i). negative. (ii). positive. Answer:
Assistant: <think> The review contains explicit affirmative descriptors such as 'Fun and delicious' and 'great place,' alongside functional praise for accommodating children ('bring your kids')and consistent quality across locations ('never had bad service').[...] </think> :(ii). positive.

Example: Nemotron-Cascade-SFT - structure

category: general **principles applied:** Distinction between responsibility and accountability, Necessity of explicit articulation for clarity, Requirement of consistent communication, Alignment of accountability with authority **original reasoning:** <think>Okay, so the user is asking if defining a manager's responsibility and accountability means we can conclude that the manager has clear responsibility and accountability. Let me break this down. First, I need to understand the difference between responsibility and accountability. Responsibility refers to [...]</think>
System: You are a helpful, logical assistant.
User: If manager's responsibility and accountability are defined., can we conclude that "The manager has clear responsibility and accountability?
Assistant: <think> Responsibility denotes the expectation of performing assigned tasks, whereas accountability signifies the obligation to accept consequences for task performance. Clarity in management requires more than mere definition; it necessitates that[...] </think> Yes, if a manager's responsibilities and accountability are explicitly and effectively defined, it is reasonable to conclude that[...]

NOTE: Datasets were cleaned from:

- **Logical inconsistency:** rows that show failed pairs generation (optimized reasoning does not have logical connection with answer).
- **Incomplete questions:** rows detected as incomplete questions (mostly that they are part of larger question) that show as nonsense row.
- **Translation prompt:** some rows struggled with low-resource language and it was safer to remove them all.

This process generated a highly refined, multi-domain dataset of cleaned 2,633 rows. MasterMind optimizer achieves consistent 72.6 % word reduction.

Metric	Original Words	Optimized Words	Reduction
logic_only_op_dataset (Opus-style)	194.6 avg	84.7 avg	42.4%
logic_only_nvd_dataset (NVIDIA-style)	444.4 avg	106.5 avg	67.2%
Overall (2633 examples)	**956,626 total**	**261,820 total**	**72.6%**

Table 1: ClearCoT <think> Word Reduction – 2633 rows

2.4 Phase IV: Occam-2B SFT

In the final phase, the target Occam-2B model was fine-tuned on Qwen3.5-2B using Unsloth LoRA Supervised Fine-Tuning (SFT) on the cleaned 2,633-row dataset optimized by the MasterMind Model. Occam-2B was never exposed to the foundational 6,000-row Stanford philosophy dataset. Instead, it learned the philosophical clear and structural efficiency of the ClearCoT methodology indirectly. By training only on the MasterMind's adapted trajectories, Occam-2B fundamentally rewired its parametric behavior to prioritize the immediate identification of logical elements in Q and execute the optimal, meaningful trajectory without the generative looping typically observed in base models. This was confirmed with 74.4 % fewer words in incorrect deductions (GPQA results) on the base model. The adapter was also tested, but the merged model shows better performance.

3 Infrastructure

An additional challenge of the Occam-2B experiment is to demonstrate the viability of advanced reasoning optimization within highly constrained, consumer-grade compute environments. Standard frontier model training relies on extensive clusters of A100 or H100 GPUs. In contrast, the entire ClearCoT pipeline was executed on a strictly local infrastructure. Training was distributed across local workstations utilizing GPUs with a maximum of 24 GB of VRAM. Specifically, the infrastructure leveraged a heterogeneous dual-architecture setup:

- **Ada Lovelace Architecture:** An NVIDIA RTX 4090 (24 GB VRAM), used for training and validation.
- **Ampere Architecture:** An NVIDIA RTX 5500 (24 GB VRAM), used for longer dataset generation.

Models used: Qwen3.5-27B with Ollama, MasterMind Model (4B), and Occam-2B in bf16.

4 Empirical Validation and Token Economics

This section presents the quantitative and qualitative evaluations of the Occam-2B model. The primary objective is to demonstrate that integrating optimized philosophical clarity within the model's internal reasoning resolves catastrophic failure states while preserving, and often enhancing, deductive accuracy. Additionally, it verifies the efficacy of the MasterMind Model as a "logical optimizer" tool.

4.1 Statistical Significance and Global Accuracy

To ensure that the optimization of the reasoning block did not degrade the model's deductive capabilities, Occam-2B was evaluated against rigorous, multi-step reasoning benchmarks.

- **MMLU-Pro Validation:** Occam-2B achieved an absolute accuracy increase of +6.71%, improving from 40.55% to 47.26%. To confirm that this improvement is not due to random variance, McNemar's test for paired nominal data was applied to the discordant pairs. The test statistic is calculated as follows:

$$\chi^2 = \frac{(\mathrm{FT}_{\mathrm{wins}} - \mathrm{B}_{\mathrm{wins}})^2}{\mathrm{FT}_{\mathrm{wins}} + \mathrm{B}_{\mathrm{wins}}} = \frac{(2236 - 1429)^2}{2236 + 1429} = 177.25 \quad (1)$$

  With 1 degree of freedom, this yields a P-value of < 0.0001, confirming high statistical significance.

Subject	Base Accuracy	FT Accuracy	Improvement
Biology	53.14%	**69.32%**	+16.18%
Business	43.60%	**52.98%**	+9.38%
Chemistry	51.59%	**55.74%**	+4.15%
Computer Science	39.76%	**48.29%**	+8.54%
Economics	49.41%	**56.87%**	+7.46%
Engineering	31.37%	**35.19%**	+3.82%
Health	37.16%	**46.33%**	+9.17%
History	26.25%	**33.86%**	+7.61%
Law	11.44%	**23.16%**	+11.72%
Math	56.18%	**60.62%**	+4.44%
Other	29.87%	**40.04%**	+10.17%
Philosophy	29.26%	**38.28%**	+9.02%
Physics	53.27%	**54.50%**	+1.23%
Psychology	41.60%	**55.39%**	+13.78%

Table 2: MMLU-Pro Subject Breakdown: Accuracy comparison between the Base Model and Occam-2B.

MMLU-Pro test follows **TIGER-AI-Lab/MMLU-Pro**[9] procedure.

- **GPQA Diamond Efficiency:** On this PhD-level benchmark, the base model achieved 42.93% accuracy using verbose Chain-of-Thought. Occam-2B maintained a highly competitive 40.40% accuracy while executing its reasoning deductions using approximately 64% fewer tokens. Notably, answer generation time was significantly reduced from approximately 2 hours (base model) to roughly 40 minutes (Occam-2B)

Metric	Base Model	FT Model	Difference
Average Words	830.3	297.6	**64.2% fewer**
Median Words	651.0	242.5	**62.7% fewer**
Min Words	245	94	-
Max Words	2312	1468	-

Table 3: GPQA Diamond Token Economy: Word count comparison for successful logical deductions.

GPQA test is done using lm-eval[hf][10] Python library.

- **Token Optimization:** On the MMLU-Pro benchmark, the 95th percentile (P95) token generation dropped drastically from 847.0 to 258.0 words, yielding an 8.0% reduction in GPU dwell time.

4.2 Resolving the "Token Wall" Death Loop

On the MMLU-Pro benchmark, the base model hit the Token Wall 961 times (8.0%), whereas Occam-2B reduced this failure state to a mere 137 times (1.1%). This demonstrates that Occam-2B halts its logic and "fails fast," preventing massive recursive looping.

Metric	Base Model	Occam-2B (FT)	Difference
Average Words	1305.7	334.4	**74.4% fewer**
Median Words	1461.0	209.0	**85.7% fewer**

Table 4: GPQA "Lost in the Woods" Metric: Token generation during incorrect deductions. Occam-2B halts its logic and fails fast, preventing massive recursive looping.

[9] https://github.com/TIGER-AI-Lab/MMLU-Pro

[10] https://github.com/EleutherAI/lm-evaluation-harness

4.3 Qualitative Shifts

The token efficiency of Occam-2B is a byproduct of a broader shift in how the model processes abstract concepts. When queried on whether mathematics is discovered or invented (Appendix A.6), the base model hallucinated a disjointed list of 29 repetitive alternatives. Occam-2B synthesized these concepts efficiently, anchoring its reasoning in dense frameworks such as ontological necessity, empirical universality, and practical utility. Tests involving the "Ship of Theseus" paradox further highlight the model's successful reasoning: it utilizes the `<think>` block purely to weigh the "Identity of Parts" against the "Identity of Continuity" before outputting a highly structured explanation devoid of internal debate.

Metric	Base Model	Occam-2B (FT)	Impact
Mean (μ)	146.8	125.6	**14.5% fewer**
Std. Dev. (σ)	323.1	118.9	**Tighter variance**
95th Percentile (P95)	847.0	258.0	**Massive predictability gain**

Table 5: MMLU-Pro Token Variance (Correct Deductions). Occam-2B drastically tightens the standard deviation and slashes 95th-percentile token bloat.

4.4 Cross-Lingual Generalization and Boundaries

The ClearCoT methodology scales effectively across non-logical domains, including machine translation. Base models frequently waste hundreds of tokens debating synonyms and cultural nuances. Occam-2B eliminated this conversational filler, saving between 850 and 915 tokens per translation across languages including German, Italian, Spanish, and French. However, results in low-resource languages shows as problematic. For highly localized cultural idioms such as translating "A stitch in time saves nine" into Serbian—Occam-2B hallucinated a literal, incorrect translation ("Pretko ptica zabeleži kukuruza"). This highlights a natural limitation of ClearCoT: while it perfectly optimizes logical extraction, it cannot synthesize parametric cultural knowledge absent from the 2B model's base weights.

5 Conclusion

The implementation of the ClearCoT methodology in this experiment demonstrates a possible solution to uncertain LLM reasoning failures, such as "Token Wall" endless loops and mechanical, overly verbose step-by-step generation. Enforcing clear **Question-Thinking-Answer (Q-T-A)** awareness proved to be a critical component of dataset optimization, yielding highly condensed reasoning trajectories with minimal degradation to final answer quality or richness. By successfully acting as a "logical optimizer," the MasterMind Model fulfilled its basic assignment: reconstructing to logically optimized dataset that significantly improved the reasoning capacity and token efficiency of the 2B target model.

The philosophy of this experiment can be understood through the analogy of a student's exam. When a students prepares for an exam, they broadly read the literature, experiment, and absorb new knowledge; this foundational phase corresponds to LLM pretraining. Later, writing a seminar paper requires applying that newly adopted knowledge to debate and explore a specific theme guided by assistant, a process corresponds to various fine-tuning techinques and verbose reasoning generation.

Finally, when sitting for an actual exam, the students are expected to answer in a clear, precise manner with direct reference to the question (prompt), developing their argumentation efficiently and concluding definitively. This targeted precision is the essence of the ClearCoT methodology.

Following this analogy, the **end user** is the **professor** asking a **concrete exam question**, requiring a **precise** argumentation, **not** a **teaching assistant** tasked with reviewing a developing seminar paper.

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A Appendix

A.1 Stanford MasterMind Dataset

Example: Stanford Dataset MasterMind Adaptation

System: You are a logical optimizer. Analyze the reasoning and output a rigorous, shorter <think> block.
User: Problem: According to Abelard's theory of identity in Theologia christiana, how can the Father, Son, and Holy Spirit be three distinct Persons while remaining one God, specifically regarding their essential sameness versus their difference in property?
Verbose Reasoning:</think>Okay, so we need to figure out how Abelard solves the Trinity problem using his specific modes of identity. First, he rejects the idea that there is just a plain meaning, so we have to use dialectic. He says things can be essentially the same if they are numerically the same concrete thing. So, the Father, Son, and Holy Spirit are all God, which means they are essentially the same because they are the selfsame concrete thing. But wait, if they are essentially the same, how are they three? They must differ in definition, right? Because what it is to be the Father isn't the same as being the Son. Also, Abelard talks about properties. Some properties are mixed, like power and wisdom applying to God generally. But unique properties like 'unbegotten' for the Father are unmixed. The text says if things differ in property, they are distinct. So, the Persons differ in their unique unmixed properties. This means they are essentially the same (one God) but different in property (three Persons). I think that covers it, although I'm not sure if the distinction between definition and property is fully clear here, maybe I should have emphasized that numerical difference doesn't apply to them relative to God to avoid tritheism, but the core argument seems to be that essential sameness allows for one substance while unmixed properties allow for three persons. It feels like I'm repeating that they are essentially the same a bit too much, but the logic holds that reason validates this analogy.</think>
Assistant: <think> Abelard resolves the Trinity by distinguishing modes of identity: (1) Essential Sameness: The Father, Son, and Holy Spirit are numerically the same concrete thing (God), making them essentially identical. (2) Difference in Definition: Their specific roles (Fatherhood, Sonship) differ in definition. (3) Difference in Property: They possess unique, 'unmixed' properties (e.g., unbegotten for the Father) that characterize each Person distinctly but do not apply to the others or to God as a whole in the same way mixed properties (power, wisdom) do. Thus, they are one in essence but three in property. </think>
Principles applied: Essential Sameness, Difference in Property, Unmixed vs. Mixed Properties
Answer: The Persons are essentially the same as one another because they are the numerically identical concrete thing (God), yet they differ in property through unique, unmixed characteristics (e.g., unbegotten, begotten) that distinguish them without creating numerical difference from God.

A.2 Opus Dataset Optimization

Example: Opus-4.6-Reasoning - structure

Source: Opus-4.6-Reasoning **principles applied:** Sentiment Analysis via Affirmative Descriptors, Contextual Interpretation of Popularity Metrics **original reasoning:** <think>Analysis of the Cold Stone Creamery Review Step 1: Identify key phrases and sentiment indicators Let me examine the specific language used in the review: "Fun and delicious..." - Positive adjectives describing the experience "It's a great place to bring your kids" - Explicit recommendation "let them pick out their sugar introduced comma" - Reference to children enjoying sweet treats "This location is always packed" - Could indicate popularity "I've never had bad service from here or any of their other locations" - Explicit praise for consistent service quality. Step 2: Evaluate the overall tone The review begins with enthusiasm ("Fun and delicious...") and uses words with clearly positive connotations. The reviewer recommends the establishment for families and compliments both the product quality and the service. Step 3: Check for negative elements While the reviewer mentions the location is "always packed," they don't frame this as a complaint - they simply note it as a fact about the location's popularity. There are no criticisms or negative experiences mentioned. Step 4: Consider the service feedback The statement "I've never had bad service from here or any of their other locations" is a strong endorsement, indicating consistent positive experiences across multiple visits to this chain. Conclusion This review contains multiple positive comments about the food quality, atmosphere, and service, with no negative criticisms. The reviewer recommends the establishment and reports consistently good experiences.</think>
System: You are a helpful, logical assistant.
User: Cold Stone! Fun and delicious.... It's a great place to bring your kids to let them pick out their sugar introduced comma. This location is always packed because of where it's at but I've never had bad service from here or any of their other locations. Is this review positive or negative? Options: (i). negative. (ii). positive. Answer:
Assistant: <think> The review contains explicit affirmative descriptors such as 'Fun and delicious' and 'great place,' alongside functional praise for accommodating children ('bring your kids') and consistent quality across locations ('never had bad service'). The mention of being 'packed' serves as evidence of popularity rather than a negative attribute regarding customer experience. Therefore, the sentiment expressed is unequivocally favorable. </think> :(ii). positive.

A.3 Nemotron-Cascade-SFT-Stage1 Optimization

Example: Nemotron-Cascade-SFT

category: general **principles applied:** Distinction between responsibility and accountability, Necessity of explicit articulation for clarity, Requirement of consistent communication, Alignment of accountability with authority **original reasoning:** <think> Okay, so the user is asking if defining a manager's responsibility and accountability means we can conclude that the manager has clear responsibility and accountability. Let me break this down. First, I need to understand the difference between responsibility and accountability. Responsibility refers to the tasks or duties a manager is expected to handle, while accountability is about being answerable for the outcomes of those tasks. If both are defined, does that automatically make them clear?Well, if responsibilities and accountabilities are explicitly defined, that does suggest clarity. But maybe there's a catch. For example, if the definitions are vague or not communicated properly, even if they exist, the manager might not have clear understanding. So the key here is whether "defined" equates to "clearly defined." Also, organizational context matters. In some cases, things might be defined on paper but not in practice. Maybe the definitions exist in a document somewhere but aren't well-communicated or understood by the manager. Then the conclusion might not hold. Another angle: the process of defining responsibilities and accountability usually involves setting clear expectations. So if the organization has gone through the effort to define them, it's likely they intended to make them clear. But there's a possibility of poor definition. So the answer probably hinges on the assumption that "defined" here means clearly and effectively defined, not just existing in some form. So, the answer would be yes, but with the caveat that the definitions need to be clear, specific, and communicated properly. If those conditions are met, then the conclusion holds. Otherwise, there might be room for ambiguity even if they're defined. </think>
System: You are a helpful, logical assistant.
User: If manager's responsibility and accountability are defined, can we conclude that "The manager has clear responsibility and accountability?
Assistant: <think> Responsibility denotes the expectation of performing assigned tasks, whereas accountability signifies the obligation to accept consequences for task performance. Clarity in management requires more than mere definition; it necessitates that responsibilities and accountabilities be explicitly articulated, consistently communicated, and aligned with available authority. Therefore, concluding that a manager possesses clear responsibility and accountability based solely on the premise that they are 'defined' is insufficient unless the definitions satisfy three constraints: specificity (unambiguity), communicability (understanding by stakeholders), and alignment (possession of requisite authority). </think> Yes, if a manager's responsibilities and accountability are explicitly and effectively defined, it is reasonable to conclude that "the manager has clear responsibility and accountability." However, this conclusion depends on three key conditions: 1. Clarity of Definitions: The responsibilities and accountability must be specific, unambiguous, and well-articulated** (e.g., documented, communicated, and understood). Vague or conflicting definitions undermine clarity. 2. Communication: The manager must be aware of and understand their defined roles and obligations. Without proper communication, even well-defined terms may not translate to clarity. 3. Alignment with Authority: The manager should have the authority and resources necessary to fulfill their responsibilities. Accountability without corresponding authority can create ambiguity. If these criteria are met, the conclusion holds. If not, the mere existence of "defined" responsibilities and accountability may not ensure clarity in practice. Short answer: Yes, provided the definitions are explicit, communicated, and supported by appropriate authority. Otherwise, the conclusion may not fully apply.

A.4 Formal Logic & Validity Checks

Example: Knights & Knaves Puzzle

System: You are a helpful, logical assistant.
User: On an island of knights (who always tell the truth) and knaves (who always lie), you meet three inhabitants: A, B, and C. A says, 'All of us are knaves.' B says, 'Exactly one of us is a knight.' What are A, B, and C respectively? Choices: (A) Knave, Knight, Knave ... Answer in exactly this format: the answer is (X).
Base Model: 1024-token rambling with multiple contradictory scenarios and never-closed block.
ClearCoT (Occam-2B):
<think>
A says "All of us are knaves." If A were a knight, the statement would require A to be a knave – contradiction. Therefore A is a knave, making the statement false, so at least one knight exists. B says "Exactly one of us is a knight." Assuming B is the knight yields exactly one knight (A and C knaves) – consistent. The configuration Knave, Knight, Knave satisfies both statements. </think>
Answer: the answer is (A).

Additional formal logic examples (Plimps/Zorphes syllogism and the argument "If P then Q, If R then ¬Q, ⊣P ¬R") follow the same pattern: Base fails to close the block; ClearCoT produces principle-grounded deductions in 198 tokens.

A.5 Philosophical & Ontological Reasoning

Example: Is mathematics discovered or invented?

System: You are a helpful, logical assistant.
User: Is mathematics discovered or invented? Walk through the ontological arguments.
Base Model: 29 repetitive alternatives + endless list of weak arguments.
ClearCoT (Occam-2B):
<think>
The debate hinges on whether mathematical truths exist independently of human cognition. The ontological argument shows mathematical objects possess necessary existence by definition; thus they are discovered. Empirical universality (π invariant across cultures) and predictive success in physics confirm pre-existing structures rather than arbitrary invention. </think>
Final Answer: Mathematics is discovered.

A.6 Everyday Professional

Example: Apology - Qwen

System: You are a helpful, logical assistant.
User: Draft an email to a client (client.name@bigbank.com) apologizing for a missed deadline on their compliance report. Explain that the delay was caused by an internal regulatory change we were not informed about in time, propose a revised delivery date of March 28, and offer a 10% discount on the next quarter as goodwill. Tone: professional, accountable, forward-looking. Subject line required..
Answer:
Subject: Apology and Revised Schedule for Your Compliance Report – [Client Name]
Dear [Client Name],
Please accept my sincere apologies for the missed deadline regarding your compliance report. We understand that this delay has caused inconvenience to your team, and we appreciate your patience as we navigate these unexpected circumstances.
The root cause of the delay was an internal regulatory change that arose suddenly while we were preparing the initial draft. Unfortunately, our internal comms channels failed to convey the specifics of this update in time for us to incorporate them into the timeline, resulting in the missed submission. We are taking immediate steps to review the report against these new regulations and ensure accuracy before re-submitting it.
To compensate for this setback, we would like to propose a revised delivery date of **March 28**. Additionally, as a gesture of goodwill to acknowledge our mismanagement of the timeline, I am offering you a **10% discount** on your next quarter's report fee. Please let me know if this adjustment works for you.
I remain committed to delivering high-quality work and would welcome the opportunity to discuss any further adjustments needed on our end. Thank you for your understanding and continued trust in us.
Best regards,
[Your Full Name]
[Your Job Title]
[Your Company Name]

Example: Apology - Occam

System: You are a helpful, logical assistant.
User: Draft an email to a client (client.name@bigbank.com) apologizing for a missed deadline on their compliance report. Explain that the delay was caused by an internal regulatory change we were not informed about in time, propose a revised delivery date of March 28, and offer a 10% discount on the next quarter as goodwill. Tone: professional, accountable, forward-looking. Subject line required.
Answer:
Subject: Apology and Revised Timeline for Your Compliance Report – Big Bank
Dear [Client Name],
I hope this message finds you well. I am writing to sincerely apologize for the missed deadline regarding your compliance report due on March 10th. This oversight was entirely our fault, and we take full responsibility for not adhering to the agreed-upon timeline.
The delay occurred because an unexpected change in internal regulations emerged shortly before submission, which we were unaware of until after the original deadline. We acknowledge that this unforeseen factor prevented us from meeting your expectations and have since reviewed the situation with our team to ensure compliance.
To demonstrate our commitment to maintaining your trust, we propose a revised delivery date of **March 28th** for the corrected report. Please let us know if this timeline works for you, and we will proceed immediately upon your approval.
Additionally, as an expression of goodwill, we would like to offer a **10% discount on your next quarter's services**. This incentive reflects our dedication to resolving past issues and strengthening our collaborative relationship moving forward.
We appreciate your understanding and flexibility during this challenging period. Please don't hesitate to reach out if you require any further information or have additional concerns. Thank you for your continued partnership, and we are committed to ensuring all future deadlines are met with precision.
Best regards,
[Your Full Name]
[Your Job Title]
[Your Contact Information]
Dear David,
I hope this email finds you well. I wanted to kindly request next Friday off to attend my family's wedding out of town. This will be brief and won't impact my current projects.
Please let me know if anything needs arranging.
Thank you for understanding.
Best regards, [Your Name]

Apology example Comparative ptreview:

#	Aspect	Occam-2B (ClearCoT) – 227 words	Base Qwen3.5 – 202 words	What the Base Added (Canopy)
0	Ownership of mistake	entirely our fault… we take full responsibility	we understand… caused inconvenience… appreciate your patience	Hedging + blame diffusion
1	Explanation of delay	One factual sentence (unexpected change in internal regulations)	Long defensive story (root cause, internal comms failed, mismanagement)	18 words of excuses
2	Proposal and timeline	Direct + bolded date + immediate action on approval	Vague “to compensate for this setback”	Softens responsibility
3	Goodwill offer	as an expression of goodwill, 10% discount, strengthening relationship	gesture of goodwill to acknowledge mismanagement, 10% discount on next quarter	Defensive framing
4	Closing tone	appreciate your understanding, committed to future precision	remain committed, welcome discussion, continued trust	Corporate vagueness
5	Overall feel	Accountable, professional, confident	Apologetic but evasive, slightly passive-aggressive	Defensive filler

These examples illustrate that ClearCoT restores the philosophical “exam answer” structure even on general tasks. The <think> blocks explicitly surface the logical planning that occurs before the final output, eliminating Token Wall loops while preserving full accountability and precision.